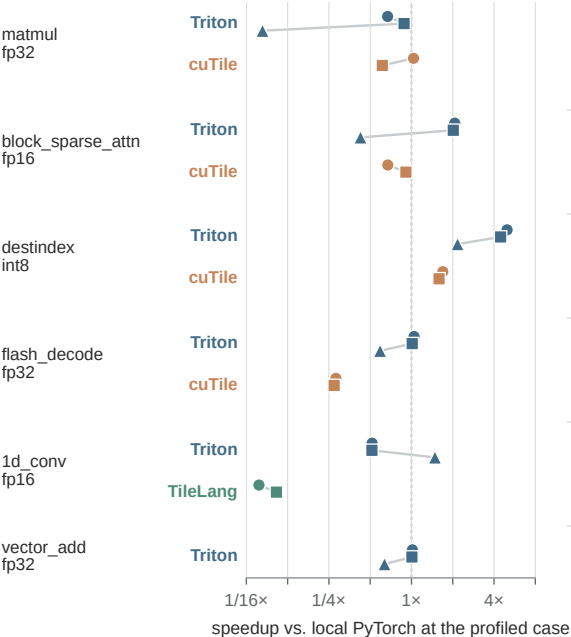


A Performance at the profiled case



B Device-native evidence (counters are not comparable across vendors)

B200	GH200	MI300X
T, C: TMA → tcgen05 TMA: C 10.7 GB, T 21.5 GB (C: larger tile)	T, C: TMA → WGMMMA (21.0M) C per WGMMMA: 1.01 STS + 1.00 LDSM (operand re-layout)	T: TensorDescriptor lowered to pointer loads (no TMA) MFMA util. 4.5%
C: tcgen05, 217 KiB smem/CTA T: legacy HMMA, 49 KiB	C: WGMMMA, 49 KiB smem/CTA T: legacy HMMA, 49 KiB	T: 256 VGPR + 256 AGPR, 20 B scratch/work-item MFMA util. 2.5%
store sectors/req: C 1.0, T 16 C: 17.3× T instructions	store sectors/req: C 1.0, T 16 C: 17.8× T instructions	T: per-element byte stores (buffer_store_byte ×4, static ISA)
16 CTAs; C: 2.1× T instr. long-scoreboard stall/issue: C 16.7, T 9.2	16 CTAs; C: 3.9× T instr. long-scoreboard stall/issue: C 5.5, T 6.2	T: 16 WGs (0.05 per CU) serial, latency-bound loop
load sectors/req: TL 17.3, T 2.3 TL occ. 6.2% (theor. 18.8%)	load sectors/req: TL 11.6, T 2.2 TL occ. 18.7% (Hopper body)	T: INT32 = 66% of VALU instr. PyTorch: MIOpen igemm + layout transposes
T: 16 sectors/req loads, DRAM 82% of peak	T: 16 sectors/req loads, DRAM 88% of peak	T: no nt loads (ATen: nt); .cg loads recover (Fig. A5)

● B200 ■ GH200 ▲ MI300X